ENGINE 4.0L

TABLE OF CONTENTS

page

ENGINE 4.0L
DESCRIPTION
DESCRIPTION - ENGINE BLOCK 71
DESCRIPTION - ENGINE 72
DIAGNOSIS AND TESTING
DIAGNOSIS AND TESTING - ENGINE DIAGNOSIS - INTRODUCTION 72
DIAGNOSIS AND TESTING - ENGINE DIAGNOSIS - PERFORMANCE 73
DIAGNOSIS AND TESTING - ENGINE DIAGNOSIS - MECHANICAL 75
DIAGNOSIS AND TESTING—CYLINDER COMPRESSION PRESSURE 77
DIAGNOSIS AND TESTING - CYLINDER COMBUSTION PRESSURE LEAKAGE 77
STANDARD PROCEDURE
STANDARD PROCEDURE - FORM-IN-PLACE GASKETS 78
STANDARD PROCEDURE - HYDROSTATIC LOCK. 79
STANDARD PROCEDURE - SERVICE ENGINE ASSEMBLY (SHORT BLOCK) 79
STANDARD PROCEDURE - REPAIR DAMAGED OR WORN THREADS 79
STANDARD PROCEDURE - ENGINE PERFORMANCE 79
STANDARD PROCEDURE - BELLHOUSING SEALANT APPLICATION 80
REMOVAL 80
INSTALLATION 82
SPECIFICATIONS
ENGINE - 4.0L 83
TORQUE - 4.0L ENGINE 87
AIR CLEANER ELEMENT
REMOVAL 88
INSTALLATION 88
AIR CLEANER HOUSING
REMOVAL 88
INSTALLATION 88
CYLINDER HEAD
DESCRIPTION 89
DIAGNOSIS AND TESTING - ENGINE CYLINDER HEAD GASKET FAILURE 89
REMOVAL 89
CLEANING 90
INSPECTION 90
INSTALLATION 90
CYLINDER HEAD COVER(S)
DESCRIPTION 91
REMOVAL 92

page

CLEANING 92
INSPECTION 92
INSTALLATION 92
INTAKE/EXHAUST VALVES & SEATS
DESCRIPTION 93
STANDARD PROCEDURE - VALVE SERVICE . . 93
REMOVAL 95
INSTALLATION 95
ROCKER ARM / ADJUSTER ASSEMBLY
DESCRIPTION 96
OPERATION 96
REMOVAL 96
CLEANING 97
INSPECTION 97
INSTALLATION 97
VALVE STEM SEALS
DESCRIPTION 97
VALVE SPRINGS
DESCRIPTION 98
STANDARD PROCEDURE - VALVE SPRING TENSION TEST 98
REMOVAL 99
INSTALLATION 99
ENGINE BLOCK
CLEANING 100
INSPECTION 100
CAMSHAFT & BEARINGS (IN BLOCK)
DESCRIPTION 101
REMOVAL
REMOVAL - CAMSHAFT BEARINGS 101
REMOVAL - CAMSHAFT 101
INSTALLATION
INSTALLATION - CAMSHAFT BEARINGS . . 102
INSTALLATION - CAMSHAFT 102
CONNECTING ROD BEARINGS
STANDARD PROCEDURE - FITTING CONNECTING ROD BEARINGS 103
CRANKSHAFT
DESCRIPTION 106
CRANKSHAFT MAIN BEARINGS
STANDARD PROCEDURE - FITTING CRANKSHAFT MAIN BEARINGS 106
REMOVAL 109
INSPECTION 110
INSTALLATION 111
CRANKSHAFT OIL SEAL - FRONT
REMOVAL 112
INSTALLATION 112
CRANKSHAFT OIL SEAL - REAR
REMOVAL 112